Quiz 10

Friday, November 20 — inverses, rank, and determinants (Week 11)

Last updated: October 12, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Strang / MIT (optional): Reading: Strang §3.5 (PDF) — rank / subspaces (Week 11)
Video: 18.06 Lec 10 — four fundamental subspaces · MIT 18.06 playlist (Strang)

Covers the prior instructional week only: Inverses, Rank, Determinants (the week of Nov 2). Midterm 2 week is skipped when choosing the prior week. Same-day Lab 5 / eigenvectors are **not** on this quiz.

Textbooks: Boyd & Vandenberghe (VMLS) · Strang §3.5 · Lachniet Octave PDF · also on Resources.

Schedule

Core

Fluency (complete prompts — practice these by hand)

C10.1. Let

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}.$$

(a) Compute A^{-1} using the 2×2 formula (or by solving $AB = I$). (b) Check $A^{-1}A = I$. (c) Solve $Ax = (4, 3)$ using $x = A^{-1}b$.

C10.2. Decide whether each matrix is invertible. Give a one-line reason (dependent columns, $Ax = 0$, or det).

(a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

C10.3. True or false, with a one-line reason: for a single right-hand side b, forming A^{-1} explicitly and then computing $A^{-1}b$ is generally preferable to solving $Ax = b$ with a solver / elimination.

C10.4. Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The RREF of A is

$$\begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

(after row swaps / scaling as needed — use this RREF). Find $\text{rank}(A)$, a basis for $C(A)$ using original columns, and a basis for $N(A)$.

C10.5. Let

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Describe $C(A)$ in words. Is $b = (1, 2, 3)$ in $C(A)$? Find a basis for $N(A)$ (or state that $N(A) = \{0\}$).

C10.6. True or false, with a one-line reason: if $\text{rank}(A) = n$ for an $m \times n$ matrix with $m \geq n$, then $Ax = 0$ has only the zero solution.

C10.7. Compute each determinant.

(a) $\det \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$

(b) $\det \begin{bmatrix} 1 & 4 & 0 \\ 0 & -2 & 5 \\ 0 & 0 & 3 \end{bmatrix}$

(c) $\det \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$

C10.8. Let $A = \begin{bmatrix} 0 & -2 \\ 1 & 0 \end{bmatrix}$. Compute $\det A$. Where do e_1 and e_2 go under $x \mapsto Ax$? What area does the unit square map to?

C10.9. True or false, with a one-line reason: if $\det A = 0$ for square A, then A is not invertible and $Ax = b$ fails to have a unique solution for every b.

C10.10. Connect the ideas. For square A, match each statement to the equivalent view (write the letter):

- (i) A is invertible
- (ii) $\operatorname{rank}(A) = n$ (full)
- (iii) $\det A \neq 0$
- (iv) $N(A) = \{0\}$

Which of (ii)–(iv) are equivalent to (i)?

C10.11 (cumulative, light). Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. What geometric map is $x \mapsto Ax$? Is A invertible? What is $\det A$?

Textbook Core

- B 11.1, 11.2, 11.4, 11.10
- S 3.5 #1 and #2 (rank / subspace portions used in class)
- L Ch. 1 Ex. 3(a), 3(b), and 3(d): determinant, rank, and inverse portions

Stretch

B 11.15, 11.25; area scaling from $|\det A|$ for a new 2×2 map (C-M2 style).

What a quiz item might change

A new 2×2 matrix, an RREF with a free variable, whether $\det A = 0$ in context, or asking for $C(A)$ vs $N(A)$ from the same matrix.